How Do We Formalize a Nonlinear Statistical Dependency Test ?

Let $\mathcal{F}$ and $\mathcal{G}$ be a characteristic RKHS

Regularization Constraint

$$\sup_{g \in \mathcal{G}} \sup_{f \in \mathcal{F}} \mathbf{Cov}(f(X), g(S)) = \langle g, \mathbb{C}\mathbf{ov} f \rangle_{\mathcal{G}} \qquad s.t \qquad \|f\|_{\mathcal{F}} = \|g\|_{\mathcal{G}} = 1$$

$\|\mathbb{C}\mathrm{ov}\|^2_{HS}$ Hilbert-Schmidt Norm of Covariance Operator Indicates Existence of Such Functions

$\|\mathbb{C}\mathrm{ov}\|^2_{HS} = 0 \iff$ Statistical Independence

The Correct Proxy To Impose Statistical Independence

How Do We Formulate Erasure Using Statistical Independence ?

$$Z_\theta = \varepsilon(X; \boldsymbol{\theta}) \quad \rightarrow$$